Toward a four-dimensional Yang–Mills construction

Notebook 02: preserving diagonal reflection positivity

Research working document

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Abstract

We give a fixed-coupling argument transferring diagonal reflection positivity to the infinite-volume limits of the original coordinate Wilson tori of side $2L^M$, with L odd. The construction uses exact Wilson slab gluing, alternating coordinate site/link cuts, local gauge-invariant layer removal, and convex separation. It requires neither uniqueness of the infinite-volume state nor estimates uniform as the coupling tends to infinity. This settles one regulator-level interface in the proposed continuum construction, subject to scrutiny of the proof below. It does not establish continuum rotations, renormalized source bounds, nontriviality, or a mass gap.

Evidence and status. This is an explicit research proof, internally checked in separate passes for the local gauge argument, geometry, slab normalization, and final limit argument; it has not received external mathematical review. The comparison architecture is adapted from the QED3 material in the pinned repository, rather than claimed as a new general method. The Wilson and odd-L details are supplied below. Notebook 01 retains the other construction obligations.

1 Precise statement and order of limits

On independent positively oriented links of $\mathbb{Z}^4$ put $U_e \in SU(2)$, with $U_{\bar{e}} = U_e^{-1}$. Haar measures have mass one. At fixed $0 < \beta < \infty$, write

$$k_\beta(g) = \exp\{(\beta/2)\operatorname{Tr} g\}, \qquad Z_\Gamma[F] = \int F(U) \prod_p k_\beta(U_p) \prod_e dU_e. \tag{1.1}$$

Every unoriented elementary plaquette occurs once. Let $C_n = \mathbb{Z}^4/n\mathbb{Z}^4$, $n = 2L^M$, for a fixed odd integer $L \geq 3$. The normalized Wilson measure is denoted μ_{C_n}. Small tori with degenerate plaquette identifications can be discarded.

A lattice reflection acts on oriented links and hence on configurations. For its ordinary pullback $rF = F \circ r$, put $\Theta F = \overline{rF}$. Reflection positivity (RP) means

$$\nu((\Theta F)F) \geq 0 \tag{1.2}$$

for every continuous local F supported in the closed positive half. Equivalent finite sums follow by replacing F with their sum.

Theorem 1.1. *Every locally weak subsequential limit ν of μ_{C_n} is reflection invariant and reflection positive across $x_1 - x_2 = 0$. By its lattice translations and signed coordinate permutations, the same holds at every lattice-preserving diagonal mirror $x_i \pm x_j = c$, $i \neq j$, $c \in \mathbb{Z}$. The original coordinate RP properties are retained as well.*

The proof constructs, for each separating observable O, auxiliary tilted measures. Its limits are strictly ordered:

$$\text{fix } \beta, O, \text{ and all cell sizes;} \quad n \longrightarrow \infty; \quad t \downarrow 0; \quad \text{then approximate local observables.} \tag{1.3}$$

No interchange with $\beta \to \infty$ is made. Applying the theorem at each lattice spacing is compatible with a volume-first continuum construction. Arbitrary simultaneous volume/cutoff limits would need additional estimates.

2 Exact Wilson slabs and reflected cell inequalities

2.1 Diagonal boundary variables

Take height $s = x_1 - x_2$ and an integer mirror $s = c$. Links in directions 1 and 2 end on this mirror or avoid it; their interiors never cross it. Links in directions 3 and 4 can lie in it. A $(1, 2)$ plaquette based at height b has heights $b, b+1, b-1, b$ and crosses the mirror exactly when $b = c$. It splits into two length-two paths between the same boundary vertices. Plaquettes in directions $(1, \mu)$ or $(2, \mu)$, $\mu = 3, 4$, have height range one and lie on one side. A $(3, 4)$ plaquette can lie wholly in the mirror. For $s = x_1 + x_2$ the crossing $(1, 2)$ plaquette is centered at $c = b + 1$; the same classification holds.

For completeness the following feature construction works for $SU(N)$. Let $u(A)$ be the real vector of real and imaginary matrix entries, so $u(A) \cdot u(B) = \operatorname{Re}\operatorname{Tr}(AB^\dagger)$ and $|u(A)|^2 = N$. Set $c_\beta = \beta/N$ and

$$\Phi_\alpha(A) = \left(\frac{c_\beta^{|\alpha|}}{\alpha!}\right)^{1/2} u(A)^\alpha, \qquad \alpha \in \mathbb{N}^{2N^2}.$$

Then

$$\langle \Phi(B), \Phi(A) \rangle = e^{c_\beta \operatorname{Re}\operatorname{Tr}(AB^\dagger)}, \qquad \|\Phi(A)\|^2 = e^\beta. \tag{2.1}$$

The coefficients are real and the series converges absolutely by Cauchy–Schwarz. If A_+, A_- are the two half-path holonomies with common initial and final vertices, their feature contraction is exactly the crossing plaquette weight.

Work on a cylinder with finite tangential period lattice $\Gamma_\parallel \subset \{s = 0\}$. At $s = c$ take boundary Hilbert space

$$\mathcal{H}_c = L^2(SU(2)^{E_c}, db) \widehat{\otimes} \bigotimes_{p \in P_c^\times} \mathcal{F}_p, \tag{2.2}$$

where E_c are links in the sheet, $P_c^\times$ the crossing plaquettes, and $\mathcal{F}$ the complexified feature space above. Give each neighboring slab a square root of the product $W_c(b)$ of weights of plaquettes contained in the sheet.

For a bounded function F strictly inside a slab $[c_0, c_1]$, define $S[F] : \mathcal{H}_{c_0} \to \mathcal{H}_{c_1}$ by integrating its interior links, multiplying every one-sided plaquette weight, the two boundary halfweights, and the feature coefficient for each inside half-path at either boundary. More explicitly, its kernel is

$$\sqrt{W_{c_1}(b_1) W_{c_0}(b_0)} \int F \prod_{p \text{ assigned to slab}} k_\beta(U_p) \prod_{p \text{ at boundaries}} \Phi_{\alpha_p}(A_{p,\text{in}}) \prod_{e \text{ in slab}} dU_e. \tag{2.3}$$

Widths are at least four, as they are in every application here. The constant feature norm and bounded finite-volume Wilson weights, followed by Minkowski's inequality, prove that (2.3) is a Hilbert–Schmidt kernel, also for complex F.

Lemma 2.1 (Gluing and adjoints)**.** *Adjacent slab kernels compose to the kernel of their union with the product of their integrands. Reflection of a slab takes its operator adjoint, with the natural boundary translation identification. No extra normalization factors occur.*

Proof. At a common sheet, kernel composition integrates each boundary link once. Feature contraction restores each crossing plaquette by (2.1); the two halfweights multiply to W_c. Every other link or plaquette already belongs to exactly one slab. All Haar measures have mass one. Reflection exchanges the two half-paths and kernel arguments; the real feature coefficients give the complex adjoint. If the width is b, reflection in its upper mirror maps its lower boundary to the translate by $q = bv$, where $v = e_1 - e_2$ or $e_1 + e_2$. Thus, with the translation unitary U_q, $S[\Theta F] = U_q S[F]^*$. This uses an integer translation even when b is odd; the two original boundary spaces need not be identified by a normal translation of length $b/2$. □

If $\Gamma = \Gamma_\parallel \oplus \mathbb{Z}\gamma$, $s(\gamma) = p$, the same graph gluing gives

$$Z_\Gamma[F_1 \cdots F_J] = \mathrm{Tr}(U_\gamma^{-1} S[F_J] \cdots S[F_1]). \tag{2.4}$$

The closing translation may have a tangential component. It is a unitary relabeling, not a new action at the seam.

2.2 Even folds and four successive directions

Suppose $p = Jb$ with J even. The straightened period is $\gamma_0 = (p/2)v \in \mathbb{Z}^4$. Denote by $\mathrm{Fold}_s F$ the product of successive antilinear reflected copies of one slab function, on the straightened torus Γ_0. Each reflected pair contributes a translate of $S[F]^* S[F]$. Their translation unitaries telescope to U_{γ_0} and cancel the closing unitary in (2.4). Hence

$$Z_{\Gamma_0}[\mathrm{Fold}_s F] = \|S[F]\|_J^J \geq 0, \qquad |Z_\Gamma[\textstyle\prod_j F_j]| \leq \prod_j Z_{\Gamma_0}[\mathrm{Fold}_s F_j]^{1/J}. \tag{2.5}$$

The second inequality is Schatten Hölder. Different boundary Hilbert spaces are allowed; the operators can be viewed as cyclic blocks in their orthogonal direct sum.

Coordinate versions, including unlike site/link boundaries, are proved in Section 3. For four orthogonal families with even successive counts $J_1, \ldots, J_4$, apply (2.5) one direction at a time. Reflections in one direction preserve the other heights. After each completed gluing one has an ordinary Wilson action, so there is no simultaneous assignment of corner halfweights. The terminal factor is the full reflected array of one original cell. Writing $N_{\mathrm{cell}} = \prod_i J_i$, this proves

$$|Z_\Gamma[\textstyle\prod_c F_c]| \leq \displaystyle\prod_c Z_{\Gamma_{\mathrm{final}}}[\mathrm{Fold}_{\mathrm{cells}} F_c]^{1/N_{\mathrm{cell}}}. \tag{2.6}$$

Its terminal factors are nonnegative by the last use of (2.5). All cell integrands used below have support strictly inside their cells.

The same boundary factorization with two reflected halves proves RP on a straight diagonal torus, including bounded positive source weights that split into reflected half products. Reflection invariance follows directly from its graph symmetry.

3 Coordinate physical operators and the insertion bound

On a finite spatial three-torus with edge set E, let $\mathcal{H} = L^2(SU(2)^E)$ and let P be Haar averaging over the spatial endpoint gauge group. Write $\mathcal{H}_{\mathrm{phys}} = P\mathcal{H}$. Let $C = \bigotimes_{e \in E} C_\beta$ be electric convolution with $k_\beta(vu^{-1})$, and let V multiply by all spatial plaquette weights. They commute with P. The positive convolution eigenvalues are $2I_{2j+1}(\beta)/\beta$; their rapid decay makes C trace class.

On the physical space define

$$B = V^{1/2} C^{1/2}, \quad A_l = B^* B = C^{1/2} V C^{1/2}, \quad A_s = B B^* = V^{1/2} C V^{1/2}.$$

Temporal-link Haar integration gives CP at every transition: its plaquette is $\mathrm{Tr}(U_e a_y U_e'^{-1} a_x^{-1})$. Consequently the ordinary coordinate partition function is exactly

$$Z_\Gamma(N) = \mathrm{Tr}_{\mathcal{H}}(CPV)^N = \mathrm{Tr}_{\mathrm{phys}}\, A_l^N = \mathrm{Tr}_{\mathrm{phys}}\, A_s^N. \tag{3.1}$$

There is no gauge-volume factor. A link cut has a factor $C^{1/2}$ on either side; a site cut splits V into $V^{1/2}$. Halfsteps are B and B^*. Thus a slab can have unlike boundary types.

Lemma 3.1 (Bounded sources in coordinate slabs)**.** *Let a slab have length $\ell \in \frac{1}{2}\mathbb{N}$ and bounded real gauge-invariant source R, $\|R\|_\infty \le M$, whose support crosses neither cut. For every even integer $p \ge 2$, its boundary operator satisfies*

$$\|S_{R,t}\|_p \le e^{|t|M} Z_\Gamma(p\ell)^{1/p}. \tag{3.2}$$

If J is even, $\sum_j \ell_j = N$, and bounded operators X_j are placed at the intervening cuts, the unnormalized insertion obeys

$$|\mathrm{Tr}(X_J S_{R_J,t} \cdots X_1 S_{R_1,t})| \le e^{|t|\sum_j M_j} \prod_j \|X_j\| \prod_j Z_\Gamma(J\ell_j)^{1/J}. \tag{3.3}$$

Proof. Glue p alternating copies of the slab and its adjoint. At link joins the factors give $C^{1/2}PC^{1/2} = CP$; at site joins they give $V^{1/2}PV^{1/2} = VP$, with that projector absorbed by adjacent physical temporal cores. The trace of $(S^*S)^{p/2}$ is the ordinary Wilson integral on length $p\ell$ with p bounded source copies. Pointwise bounding their product gives (3.2). Finite representation truncations or the bounded glued integral justify the kernel limit. No pointwise positivity of the kernel of $C^{1/2}$ is asserted. Schatten Hölder gives (3.3). □

For general bounded complex gauge-invariant slab integrands the operator is defined linearly: split real and imaginary parts, and write a bounded real function as a difference of two strictly positive bounded functions, each an exponential of a bounded real function. The kernels glue linearly and reflection is the adjoint. This proves the coordinate version of (2.5) for half-integer widths and even counts. A tangential closing shift acts unitarily on both $\mathcal{H}$ and $\mathcal{H}_{\mathrm{phys}}$ and gives the same twisted-trace formula.

For a selected finite set of spatial links S at a link cut, replace the electric kernel K_S by that of $C_S^{1/2} X C_S^{1/2}$ and divide by $K_S > 0$. Include temporal parallel transports at its endpoints. The resulting local insertion is denoted G_X; $G_I = 1$. Throughout the rest of this note, X is restricted to finite Peter–Weyl kernel operators and their unitization that commute with endpoint gauge transformations. Their insertion kernels and G_X are continuous, $X \otimes I$ commutes with P, and G_X is gauge invariant. This class contains the self-adjoint combinations and $I + isA$ used below. Disjoint spatial supports at a common cut give an operator norm at most the product of the individual norms. Lattice reflections and relabelings replace X by adjoints or unitary/antiunitary conjugates, preserving its norm.

4 Local gauge-invariant reduction to two slices

Lemma 4.1 (Finite derivative loss)**.** *For a finite Peter–Weyl polynomial f on $SU(2)$ of maximum spin r, $T_f = C_\beta^{-1} M_f C_\beta$ maps C^∞ continuously to itself and*

$$\|T_f\psi\|_{H^s} \le C_{\beta,f,s}\, \|\psi\|_{H^{s+2r}}. \tag{4.1}$$

The tensor version holds on any fixed finite edge set.

Proof. The spin-j convolution eigenvalue is $c_j = 2I_{2j+1}(\beta)/\beta > 0$. Multiplication by f connects only $|k-j| \le r$. In

$$I_m(\beta) = \frac{(\beta/2)^m}{m!} S_m, \qquad 1 \le S_m = \sum_{l\ge 0} \frac{(\beta^2/4)^l m!}{l!(m+l)!} \le e^{\beta^2/4},$$

ratios on that finite band grow at most as $C(1+j)^{2r}$. Each multiplication block has norm at most $\|f\|_\infty$; weighted square summation proves (4.1). Sobolev embedding and the finite tensor product give the stated smooth mapping property. □

We record the gauge step explicitly because replacing the law by a gauge-fixed one would obscure the support issue. Suppose a local gauge-invariant polynomial P has lowest coordinate height j. For its bottom spatial edges $e = (x, y)$ put

$$b_e = U_{(j,x),(j,y)}, \quad a_x = U_{(j,x),(j+1,x)}, \quad u_e = a_x^{-1} b_e a_y, \quad v_e = U_{(j+1,x),(j+1,y)}.$$

Include auxiliary a_x at all necessary endpoints. Gauge transformation $g_{j,x} = a_x^{-1}$ sets those temporal links to one and leaves every upper link unchanged. It proves the identity of observables

$$P(U) = \widetilde{P}(u, U_{\ge j+1}) = \sum_i f_i(u) P_i(v, \eta). \tag{4.2}$$

There is no change of the probability measure.

Condition on all links except the finite collection v. Its downward plaquettes have trace $\mathrm{Tr}(u_e v_e^{-1})$. Every other plaquette containing v is spatial at $j+1$ or goes up to $j+2$. Include each such complete plaquette once in $W(v;\eta)$. The density factors as

$$H(\text{exterior}) K_S(v,u) W(v;\eta).$$

W, its inverse, and all required derivatives are bounded uniformly in ambient volume at fixed β and fixed support. Since $C_S T_f = M_f C_S$ and $K_S(v,u) = K_S(u,v)$, define

$$Q(v,\eta) = W(v;\eta)^{-1} \sum_i T_{f_i}[W(\cdot;\eta) P_i(\cdot,\eta)](v). \tag{4.3}$$

Conditional integration proves $\mu(P) = \mu(Q)$ exactly. Q is smooth and reads only links at heights at least $j+1$, reaching up at most to $\max(j+2, \text{old top})$. It is gauge invariant: u_e and v_e transform by the same upper endpoint gauges, C_S commutes with their action, and $T_{hf}(hB) = h(T_f B)$. Thus (4.3) descends directly to the original upper links without reintroducing the removed links.

Now tilt by e^{tV_n}, where V_n is a fixed-spacing array of reflected copies of a smooth bounded local source. Only a uniformly bounded number of copies depend on the integrated v; call their sum R_n. It may also read lower exterior links. With those frozen, the exact tilted replacement is

$$Q_{n,t} = W^{-1} e^{-tR_n} \sum_i T_{f_i}[e^{tR_n} W P_i]. \tag{4.4}$$

Finite derivative loss and compactness give $\|Q_{n,t} - Q\|_\infty \le C_P t$ for $0 \le t \le 1$, uniformly in n. The analogous top-layer removal uses $u_e = a_x b_e a_y^{-1}$ and goes down one slice.

Proposition 4.2 (Paired two-layer reduction)**.** *For every continuous gauge-invariant local P, every integer a, and $\epsilon > 0$, there is a finite local gauge-invariant polynomial Y_ϵ supported in $[a, a+1]$, and $C_\epsilon < \infty$, with*

$$\begin{aligned} |\mu_{n,t}(P) - \mu_{n,t}(Y_\epsilon)| &\le \epsilon + C_\epsilon t, \\ |\mu_{n,t}(rP) - \mu_{n,t}(rY_\epsilon)| &\le \epsilon + C_\epsilon t, \end{aligned} \tag{4.5}$$

where r is reflection in $a+1/2$. No invariance of the tilted measure or source array under r is required. The estimates hold for all sufficiently large n, once the finite neighborhoods produced by reduction and their reflections embed without identifications.

Proof. First uniformly approximate a continuous input by a finite Peter–Weyl polynomial and average over its finite endpoint gauge group, charging this initial approximation to the total ϵ allowance. Remove bottom layers up to a, then top layers down to $a+1$. The first operations never exceed the retained top bound since $j+2 \le a+1$; the second never drop below a since $j-2 \ge a$. Each operation enlarges spatial support by finitely many plaquettes. After each step approximate its smooth output uniformly by a finite Peter–Weyl polynomial and Haar-average over the finite endpoint gauge group. The average preserves both support and finite representation content. Choose the finite list of approximation errors with total below ϵ. Expectation errors telescope, so subsequent operator norms do not amplify earlier errors. The finite sum of tilted errors is $C_\epsilon t$. Reflect the same unperturbed construction for rP; its tilted error proof needs only the same local derivative bounds, not reflection symmetry of the tilt. This proves (4.5). □

5 The two frames on the original odd-L family

Put $a = n/2 = L^M$, $e_- = e_1 - e_2$, $e_+ = e_1 + e_2$, and introduce

$$\Gamma_D = \langle ae_-, 2ae_+, ne_3, ne_4\rangle, \qquad \Gamma_T = \langle ae_1, 4ae_2, ne_3, ne_4\rangle. \tag{5.1}$$

Both covolumes are n^4. These are auxiliary tori; C_n is unchanged.

Let O be a real gauge-invariant local Peter–Weyl polynomial. Choose a source radius R_O large enough to contain its full link support and let $h = L^u > 16R_O + 16$. Fine diagonal cells have widths h in $(x_1 - x_2, x_1 + x_2, x_3, x_4)$, with integral diagonal offsets and half-integral transverse offsets. The source is strictly inside one cell; V_n is the sum of its reflected copies.

For an insertion box $B = y + [-r, r]^4$, choose fixed powers $q = L^v$, $b = qh$, $H = L^z$, with $b \mid H$, and put $w = b/2$, so

$$w \ge 2r + 6h, \qquad H/w \ge 16. \tag{5.2}$$

For large M, $h \mid b \mid H \mid a$. Both b, H are odd; w is strictly half-integral. Coordinate walls spaced by w therefore alternate between site and link cuts.

5.1 Periods and counts

On C_n the first diagonal height has period n and tangential cross-section generated by ne_+, ne_3, ne_4. Straightening replaces the closing vector ne_1 by ae_-. The subsequent height periods are $2n, n, n$. Thus the first fold ends on D_n with even counts

$$(n/h, 2n/h, n/h, n/h), \qquad N = 2n^4/h^4. \tag{5.3}$$

The same fold on D_n with width H has $N' = 2n^4/H^4$ cells. The diagonal height transformation has determinant two, explaining the cell volume $h^4/2$. Both original and adapted periods belong to the wall group's translation subgroup generated by $he_-, he_+, 2he_3, 2he_4$; all reflected arrays are therefore periodic.

For the first coordinate height on D_n, the period is a, with cross-section $\langle 4ae_2, ne_3, ne_4\rangle$ and closing vector $ae_1 - ae_2$. Its tangential shift is the unitary relabeling $-ae_2$. Straightening gives T_n. With $A = a/b$, the coordinate counts are

$$(2A, 8A, 4A, 4A), \qquad N'' = 256A^4 = n^4/w^4, \tag{5.4}$$

all even. In units of w the original period generators are $(2A,-2A),(4A,4A),4Ae_3,4Ae_4$. First reduce the first index by the first generator, then the second by $(0,8A)$, then the transverse ones. This proves that the brick (5.4) represents both D_n and T_n, including the intermediate cellwise folding geometry.

5.2 Clean planes and distinct insertion cells

The reflected source centers have at most four residues modulo h in either first coordinate, and at most two in the transverse ones. Indeed, writing $A_\pm = x_1 \pm x_2 - c_\pm$, their first coordinates are

$$x_1' = x_1 - A_-\epsilon_- - A_+\epsilon_+ + h(z_- + z_+), \quad x_2' = x_2 + A_-\epsilon_- - A_+\epsilon_+ + h(z_+ - z_-). \tag{5.5}$$

For two adjacent half-integer planes $c, c+1$, one source-center residue forbids at most $2R_O+2$ of the h half-integer classes modulo h. Hence two adjacent clean link planes exist in each coordinate direction. Here clean means that neither plane meets any source box.

Choose coarse offsets from the fine wall families so that

$$\begin{gathered} y_1 - y_2 - c_-' \in [2w, 2w+h), \quad y_1 + y_2 - c_+' \in [4w, 4w+h), \\ y_\mu - c_\mu' \in [w, w+h), \qquad\qquad \mu = 3,4. \end{gathered}$$

For the resulting coarse wall group, put $m = H/w \geq 16$, an even integer. Every image center has form

$$gy = y + w\kappa(g) + s(\epsilon), \qquad |s(\epsilon)|_\infty < 2h,$$

where

$$\begin{aligned} \kappa(g) = (&-2\epsilon_- - 4\epsilon_+, 2\epsilon_- - 4\epsilon_+, -2\epsilon_3, -2\epsilon_4) \\ &+ m(z_- + z_+, z_+ - z_-, 2z_3, 2z_4). \end{aligned} \tag{5.6}$$

The sixteen parity vectors are distinct; coordinate differences are at most six. Since $m \geq 16$, equality of indices forces equality of the parity vectors and then of every z. Thus $g \mapsto \kappa(g)$ is injective.

Coordinate walls of width w have residues $o, o+h/2$ modulo h. Choose o modulo $h/2$ on the half-integer mesh. A source residue excludes at most $4R_O+2$ classes; all four exclude at most $16R_O+8 < h$. An allowed representative can be selected in

$$(y_\mu + r + 2h - w,\ y_\mu - r - 2h),$$

whose length is at least $2h$. Equations (5.2) and (5.6) then give

$$gB \subset o + w\kappa(g) + (0,w)^4. \tag{5.7}$$

Every source box is also strictly inside one coordinate cell. The periods in Γ_D belong to the coarse diagonal translation group because $H \mid a$. If two indices in (5.6) differ by a period, translate one group element by that period and use injectivity. They represent the same coarse orbit. Consequently exactly N' distinct coordinate cells contain amplified insertions.

After the coordinate fold of one occupied cell, all insertion cuts have the same coordinate direction. If its relative cut position is $d \in \frac{1}{2}\mathbb{Z}$, $0 < d < w$, successive gaps are $2d$ and $2(w-d)$, integers in $\{1,\dots,2w-1\}$. The residual shift in (5.6) allows no stronger universal margin; strict inclusion and this finite set of gaps suffice. The number J of cuts is one of the even counts (5.4). Each source lies wholly between consecutive operator cuts, because the original clean cut and all its coarse images avoid sources. At one cut the insertion boxes occupy disjoint spatial cells.

6 Pressure, source comparison, and amplification

6.1 The only thermodynamic input

Bounded finite-range Wilson interactions have a common pressure p_β on the coordinate and skew tori above:

$$\log Z_\Gamma[1] = p_\beta|\Gamma| + o(|\Gamma|). \tag{6.1}$$

An elementary verification is to tile a fundamental domain by fixed large coordinate boxes, deleting plaquettes at their boundaries. Each deleted plaquette changes $\log Z$ by at most β. Normalized Haar integrals factor on the disconnected boxes; leftover links have integral one. First let the torus grow, so its boundary fraction vanishes, then let the box size grow. The internal seam fraction is $O(1/\text{box size})$. Comparing boxes also proves the existence and uniqueness of their limiting pressure. The argument works for bounded-aspect skew parallelepipeds and uniformly over finitely many rectangular shapes. No quantitative convergence rate or tilted pressure is needed.

6.2 Full source arrays are essential

Use the reflected source arrays V_C, V_D in all N fine cells. Every fine-wall reflection is an integer affine signed permutation preserving the coordinate Wilson law. Thus $\mu_C(V_C) = N\mu_C(O)$, even though diagonal RP on C_n has not been assumed. Every cell factor $e^{tO^{[c]}}$ has the same complete fold e^{tV_D}; (2.6) gives

$$Z_C[e^{tV_C}] \le Z_D[e^{tV_D}]. \tag{6.2}$$

Jensen and convexity of $F_D(t) = \log(Z_D[e^{tV_D}]/Z_D[1])$ imply

$$\mu_C(O) \le \mu_{D,t}(O) + \frac{\log Z_D[1] - \log Z_C[1]}{tN}, \qquad t > 0. \tag{6.3}$$

Here $F_D'(t) = N\mu_{D,t}(O)$ because the tilted law is invariant under its wall group. Equation (6.1) makes the error tend to zero at fixed t. A single-source unnormalized partition function raised to N would give an incorrect normalization; (6.2) uses full arrays on both tori.

6.3 Amplifying a clean-plane insertion

Fix G_X at one clean coordinate link plane and use the geometry above. Coarse diagonal walls preserve V_D, so the normalized coarse chessboard consequence of (2.6) is

$$|\mu_{D,t}(G_X)|^{N'} \le \frac{Z_D[e^{tV_D}\prod_g G_X^{[g]}]}{Z_D[e^{tV_D}]}. \tag{6.4}$$

For clarity, apply (2.6) to one occupied coarse cell and the empty ones: their complete folds contain respectively the N' copies and no insertion, while each contains the same source array. This gives (6.4) with exactly the displayed denominator.

Apply the coordinate cell inequality to the numerator. If cell c contains m_c sources, its complete fold has $N''m_c$ source copies; $\sum_c m_c = N$. If occupied, it also has N'' copies of its operator, all of norm $\|X\|$. The unnormalized bound (3.3) therefore costs $e^{tN''m_c\|O\|_\infty}\,\|X\|^{N''}$ for occupied cells, and just the exponential for empty cells. In that formula the source-free factors satisfy

$$\frac{1}{J}\sum_j \log Z_\Gamma(J\ell_j) = p_\beta|\Gamma|\sum_j \ell_j + o(n^4) = p_\beta n^4 + o(n^4). \tag{6.5}$$

Uniformity follows from the fixed finite set of gap lengths and time directions. All spatial lengths and $J\ell_j$ grow in bounded aspect ratios. After the $1/N''$ cell roots, the numerator bound is

$$\|X\|^{N'} e^{tN\|O\|_\infty} e^{p_\beta n^4 + o(n^4)}.$$

The one original denominator is at least $e^{-tN\|O\|_\infty} e^{p_\beta n^4 + o(n^4)}$. Taking the N'-th root proves

$$\limsup_{n\to\infty} |\mu_{D_n,t}(G_X)| \le e^{2t(H/h)^4\|O\|_\infty} \|X\|. \tag{6.6}$$

All auxiliary sizes were fixed before this limit. In particular H may depend on the support of X, but not its norm. This also proves (6.6) for the local unitization $I + isA$.

7 Removing the tilt in one common state

Fix a subsequence $\mu_{C_n} \to \nu$ and $t_j \downarrow 0$. For each j, extract a local weak limit λ_j of μ_{D_n,t_j} along a subsequence of those same volumes. Then extract $\lambda_j \to \lambda$. Compactness follows from the countable product of compact link groups. Equation (6.3) gives $\nu(O) \le \lambda(O)$. All these laws are gauge invariant; diagonal RP and invariance at the chosen fine wall survive both limits.

For every fixed finite kernel X, (6.6) holds along every volume subsequence and yields

$$|\lambda(G_X)| \le \|X\|. \tag{7.1}$$

Thus one common λ satisfies all these inequalities; no operator-dependent choice of state is involved. Put $\mathcal{L}(X) = \lambda(G_X)$, so $\mathcal{L}(I) = 1$. For self-adjoint A, apply (7.1) to $I + isA$:

$$|1 + is\mathcal{L}(A)|^2 \le 1 + s^2 \|A\|^2.$$

Let $s \to 0$ through both signs. Then $\operatorname{Im} \mathcal{L}(A) = 0$, and linearity gives $\mathcal{L}(X^*) = \overline{\mathcal{L}(X)}$. Reflection swaps the crossing kernel arguments, hence $rG_X = \overline{G_{X^*}}$. Since λ is a probability law,

$$\lambda(rG_X) = \lambda(G_X). \tag{7.2}$$

Every smooth gauge-invariant two-layer observable Y is uniformly approximable by these insertions. After temporal parallel transport, enlarge the selected spatial set so that $Y = Y(u, v)$, put $K_Y(v, u) = K_S(v, u)Y(u, v)$, and let P_m be product Peter–Weyl cutoff. Its smooth kernels $P_m K_Y P_m$ converge uniformly to K_Y. The bounded finite-rank operator

$$X_m = C_S^{-1/2} P_m K_Y P_m C_S^{-1/2}$$

commutes with endpoint gauge transformations and has $G_{X_m} = (P_m K_Y P_m)/K_S \to Y$ uniformly. Positivity of every retained convolution eigenvalue and of K_S justifies both divisions. No uniform bound on $\|X_m\|$ is required: the tilt has already been removed for each fixed operator. Uniform approximation extends (7.2) to continuous Y.

Proposition 4.2, followed by $t \downarrow 0$ and $\epsilon \downarrow 0$, extends invariance to every local gauge-invariant P at the same clean plane. Gauge averaging a cylinder test extends it to all local functions. Applying this at two adjacent clean planes in each direction proves that λ is invariant under $2\mathbb{Z}^4$ translations. This step establishes coordinate reflection *invariance*; its diagonal RP came from the adapted tilted measures.

8 Convex separation proves the theorem

Keep the desired mirror $x_1 - x_2 = 0$ fixed. Let $\mathcal{R}$ be the set of gauge-invariant probability laws invariant and RP at that mirror, and invariant under $2\mathbb{Z}^4$. It is compact and convex: on the compact configuration space, each invariance condition is a closed linear equality and each continuous local RP test is a closed affine inequality.

Given any real gauge-invariant local Peter–Weyl polynomial P, translate it by an even vector into the positive side of the fixed mirror, choose h sufficiently large, and choose the other offsets so its full support lies strictly inside that diagonal cell. Even lattice translations form a full-rank sublattice, so this has no parity obstruction. The source construction provides a law $\lambda \in \mathcal{R}$ dominating the translated expectation. Use the original law's full translation invariance and λ's even translation invariance to return to P. Therefore

$$\nu(P) \le \sup_{\lambda \in \mathcal{R}} \lambda(P) \quad \text{for every such } P. \tag{8.1}$$

If $\nu \notin \mathcal{R}$, separation of the compact convex set would give a strictly separating continuous real function. Cylinder approximation, finite Peter–Weyl approximation, and then averaging over the finite endpoint gauge group preserve strict separation and produce a gauge-invariant finite polynomial. All expectations under the gauge-invariant laws are unchanged by that averaging. This contradicts (8.1). Hence $\nu \in \mathcal{R}$. The coordinate tori already give ν full lattice translations, signed permutations and coordinate RP. These transfer the proved diagonal statement to the other mirrors and finish Theorem 1.1.

This is averaging of a *test function* in a separation argument, not an assumption that translating or averaging RP states preserves RP at a fixed plane. No uniqueness of a Gibbs state is used.

9 What this buys for the continuum problem

The result supplies a way to retain both coordinate and diagonal RP in the same volume-first lattice theory. It repairs three concrete interfaces: local gauge descent, a normalized clean-plane operator bound, and the incompatible parity in the convenient QED3 placement. The first comparison also requires full reflected source arrays; omitting them loses the volume divisor in the pressure error.

For renormalized continuum observables, passing these inequalities requires their common convergence on the relevant reflected test algebra. That convergence and its all-order bounds remain dependent on the model-specific RG estimates. Constants in the local reduction may grow with β; this proof gives no uniform ultraviolet estimate.

The remaining main obligations are:

1. Verify the selected global four-layer complex-source chain and the upstream running-coupling/cutoff input in the deposited account.

2. Prove the interacting angular analyticity and growth estimates that would turn the two reflection directions into full rotations. Notebook 01 explains why the reflection directions alone do not do so.

3. Establish an interacting nonzero positive-energy witness, then construct and identify an adequate family of mixed gauge-invariant observables. A single scalar sector is insufficient to identify Yang–Mills theory.

4. Justify vacuum-sector reconstruction and separately prove any claimed mass gap.

No item in this list is discharged by the fixed-coupling theorem.

References

[1] *Yang–Mills LLM experiment*, public repository at commit `e392ef45013a581664bab76b90867170b30fe10e`. https://github.com/ethanwsussman/Yang--Mills-LLM-experiment/tree/e392ef45013a581664bab76b90867170b30fe10e.

[2] *QED3 account*, same repository, `QED3_slop/qed3.pdf`, Sections 3.2–3.3, printed pp. 20–25, especially (3.17)–(3.27); Section 4.4, printed pp. 38–40, especially (4.17)–(4.22). Source of the slab/comparison architecture; its fermionic theorem is not imported as a Wilson proof. https://github.com/ethanwsussman/Yang--Mills-LLM-experiment/blob/e392ef45013a581664bab76b90867170b30fe10e/QED3_slop/qed3.pdf.

[3] A. Jaffe and E. Witten, *Quantum Yang–Mills Theory*. https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf.